Cost-Aware Multi-Frame Deferral for Promptable Video Segmentation

Anonymous CVPR submission

Abstract

Promptable video segmentation models such as SAM2 propagate an object mask through a video from a few expert prompts. The propagated mask drifts under occlusion and appearance change, and repairing it requires further expert intervention. Existing frame-selection methods decide where to place these corrections, but they fail to account for the trade-off between the resulting accuracy gain and the expert cost incurred in generating those corrections. We formulate corrective prompting as a variable-cardinality learning-to-defer problem: given the current propagation and a specified expert cost, our method decides whether to request assistance, how many corrections are justified, and which frames receive them. A cardinality model estimates the number of frames to correct, and a multi-frame selection model selects the frames to be corrected conditioned on the frames already selected, so the requested prompts are complementary to one another. Experiments with SAM2.1 on four datasets covering object transformation, camouflage, colonoscopy, and thyroid ultrasound show that our method achieves the best quality–cost trade-off among learned methods. Relative to the initial propagation, it improves mIoU by 5.6%–47.8%. Compared with the strongest adapted baseline, Top-$k(x)$, it reduces corrective prompts by 3.1%–9.2% and system cost by 5.4%–11.4%, while requesting and applying the selected corrections within a single correction round. Code is released upon acceptance.

1. Introduction

Recent promptable segmentation models such as SAM2 [14] can segment objects throughout a video from a small number of mask, box, or point prompts. Propagation from these prompts can nevertheless become inaccurate under appearance change, occlusion, rapid motion, and background similarity [3]. Experts must then inspect the propagated segmentation and place corrective prompts [10]. Each inspection and each corrective prompt incurs a cost, which makes corrective prompting a human–AI decision problem: whether to request expert assistance, how many corrections are justified, and on which frames to place them.

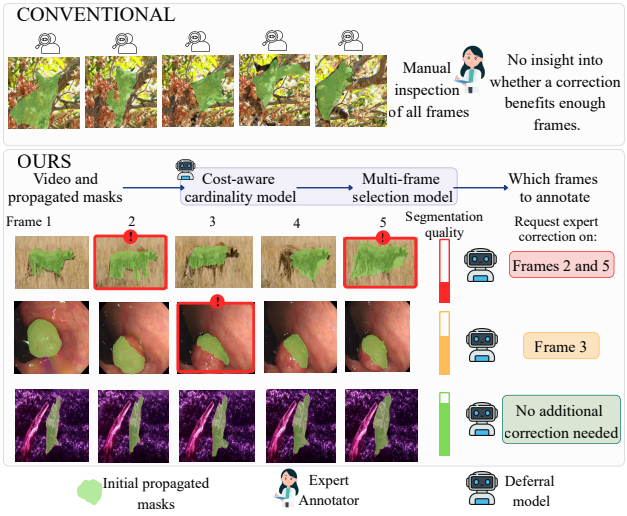


Figure 1. Conventional correction requires an expert to inspect the propagated masks and decide where to intervene, without knowing whether a correction will improve only nearby frames or benefit the video more broadly. This can underuse expert effort by spending it on corrections with limited benefit. Our framework instead accounts for intervention cost and decides whether video correction is worthwhile, how many frames should be corrected, and where those corrections should be placed.

Existing frame-selection approaches address where corrections should be placed, scoring candidate frames by uncertainty, predicted mask quality, expected improvement, or diversity [2, 18, 22]. Each candidate is scored independently, so a set assembled from the highest-scoring frames can spend several corrections on the same failure while leaving others untouched. These approaches also operate under a predefined annotation budget, fixing the number of corrections in advance rather than inferring it from the quality of the current propagation. Approaches that avoid a fixed budget instead correct one frame at a time, alternating propagation, expert review, and annotation after every intervention [18].

A related decision has been studied extensively in classification through learning to defer (L2D), which learns when expert intervention is warranted by explicitly balanc-

ing prediction quality against the cost of consulting an expert [9, 16]. Conventional formulations choose between retaining a machine prediction and replacing it with an expert one, and recent extensions also learn how many experts to consult for each input [8, 23]. L2RP [13] brings this reasoning to promptable video segmentation, deciding whether to retain the current propagation or request a correction on one selected frame. However, L2RP can request at most one correction for the entire video, which may be insufficient when propagation errors occur in multiple parts of the video. In each case, the consultations are treated as independent, but a corrective prompt re-propagates through the video, so the number of corrections worth requesting depends on which frames receive them.

We therefore formulate corrective prompting as a *variable-cardinality learning-to-defer* problem. Given the current propagated segmentation and a specified expert cost, our framework determines whether expert assistance is needed, how many corrective prompts to request, and which frames should receive them. The selected frames are constructed as a complementary set rather than ranked independently, allowing each choice to account for the frames already selected. All corrections are then requested in a single batch and incorporated through one re-propagation, avoiding the repeated expert-interaction and propagation rounds required by sequential correction methods.

Our main contributions are as follows:

- We formulate corrective prompting for promptable video segmentation as a variable-cardinality learning-to-defer problem, where the system jointly determines whether correction is needed, how many frames should be corrected, and which frames should be selected.
- We propose a framework that combines a cardinality model for determining how many corrections are warranted under a given expert cost with a multi-frame selection model for selecting the corresponding frames, with all expert prompts requested in a single correction round.
- We evaluate our framework on four datasets spanning object transformation, camouflage, colonoscopy, and thyroid ultrasound, comparing it against fixed-budget frame-selection methods and adapted learning-to-defer baselines.

2. Related work

2.1. Frame selection for corrective prompting

Frame selection for efficient human annotation has been studied in both active learning and interactive video segmentation. Active-learning approaches often prioritise frames using criteria such as uncertainty, representativeness, or diversity, with the acquired annotations subsequently used to improve the model [6, 12, 17]. For example, HVSA [11] combines uncertainty- and diversity-based selection with video-specific fine-tuning.

Interactive video segmentation considers a more directly related setting, in which user corrections are used to improve the propagated masks of the current video while the segmentation model remains fixed [20]. Within this setting, several methods have studied where and when the user should intervene. IVOS-W [22] recommends the next frame according to the expected video-level improvement after annotation, showing that the frame with the poorest current mask is not necessarily the most useful one to correct. EVA-VOS [2] selects the next frame using representations derived from the image and predicted mask, favouring frames that differ from those already annotated. Lazy-XMem [18] instead monitors uncertainty in the propagated masks and triggers a correction when the prediction becomes unreliable. Although these methods improve where the user intervenes, they select correction frames sequentially. When multiple corrections are needed for a video, this requires repeated cycles of propagation, expert review, and frame selection before all necessary corrections are identified [2]. Selecting a set of correction frames for each video can reduce this back-and-forth process while allowing the number of requested frames to adapt to the current propagation quality and the cost of expert assistance.

2.2. Learning to defer for promptable video segmentation

Learning to defer framework decides whether to retain a machine prediction or request expert assistance. Conventional formulations learn this decision by balancing the expected errors of the machine and expert against the cost of expert consultation [4, 7, 9, 16].

Related human–AI collaboration methods extend this idea to settings with multiple experts. LECODU [23] predicts how many experts should be consulted for each input and then randomly samples that number from the available expert pool before combining their predictions with the machine output. Top-k [8] learning to defer instead ranks experts according to their estimated suitability for the input and selects the highest-ranked k. Its adaptive Top-k(x) formulation additionally allows the consultation cardinality to vary across inputs.

Most L2D methods target classification. L2RP [13] is, to our knowledge, the first application of L2D to promptable video segmentation, but it defers at most once per video. Extending the setting to multiple corrective prompts is not straightforward, because each correction re-propagates through the video and changes how much the remaining candidates would add. This also prevents direct application of Top-$k(x)$ [8], which scores each expert independently and can therefore select several adjacent frames that address the same failure. Our framework instead selects the corrective frames jointly, conditioning each choice on the frames

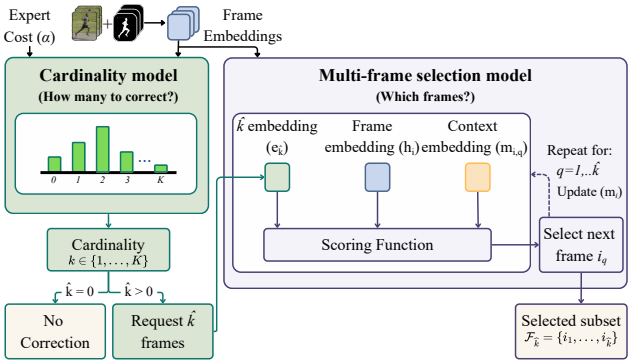


Figure 2. Overview of the proposed cost-aware multi-frame deferral framework. Given the video frame embeddings and expert cost α, the cardinality model predicts whether correction is required and, if so, the number of frames $\widehat{k}$ to prompt. Conditioned on the predicted cardinality, the multi-frame selection model constructs the correction set $\mathcal{F}_{\widehat{k}} = \{i_1, \ldots, i_{\widehat{k}}\}$ over $\widehat{k}$ selection steps, updating the context after each selected frame. The resulting frames are then requested for expert prompting in a single correction round.

already selected, and allows the number of corrections to vary with the video.

3. Method

Overview. A promptable video segmentation model requires an initial user prompt to identify the object of interest. Given this required prompt, the segmentation model propagates the corresponding object mask throughout the video to produce an initial segmentation. Our framework evaluates this propagation to determine whether additional corrective prompting is required, how many corrections should be requested, and where they should be placed. As shown in Figure 2, the framework consists of two main components: a cardinality model and a multi-frame selection model. The cardinality model selects $\widehat{k} \in \{0, \ldots, K\}$ by balancing the expected segmentation improvement from additional corrections against their expert prompting cost. If $\widehat{k} = 0$, the initial propagation is retained because the expected improvement does not justify the expert cost. Otherwise, the multi-frame selection model selects $\widehat{k}$ frames for expert annotation. The expert annotates all selected frames, and the segmentation model uses the resulting prompts together to update the segmentation throughout the video in a single propagation.

3.1. Preliminaries

3.1.1. Learning-to-Defer.

Conventional learning to defer considers a system composed of a machine predictor and a human expert. Given an input x, the machine produces a prediction $\widehat{y}^m$, while the expert can provide an alternative prediction $\widehat{y}^e$. A deferral policy $r(x) \in \{0, 1\}$ decides whether to retain the machine prediction or defer to the expert. Given the ground-truth target y, the resulting system loss is

$$\mathcal{L}_{\text{L2D}}(\mathsf{r}, x, y) = \mathbf{1}[\mathsf{r}(x) = 0]\, \ell_m(y, \widehat{y}^m) + \mathbf{1}[\mathsf{r}(x) = 1]\, [\ell_e(y, \widehat{y}^e) + \alpha]\,, \tag{1}$$

where $\ell_m(\cdot)$ is the loss incurred when retaining the machine prediction, $\ell_e(\cdot)$ is the loss associated with the expert prediction, and α denotes the cost of requesting expert input. The policy therefore learns to defer only when the expected reduction in prediction loss justifies the additional expert cost.

3.1.2. Promptable Video Segmentation Setting.

In our setting of promptable video segmentation, the machine prediction is the segmentation obtained by propagating an initial user prompt throughout a video. Expert assistance does not replace this prediction directly; instead, the expert provides corrective prompts on selected frames, which the segmentation model uses to update the video-wide segmentation.

Formally, let $v = (x_1, \ldots, x_N)$ denote a video containing N frames, with corresponding ground-truth segmentation masks $y = (y_1, \ldots, y_N)$. For each frame i, p_i denotes the expert-provided annotation that serves as the user prompt to the segmentation model. The set $\mathcal{F}_{\text{init}} \subseteq \{1, \ldots, N\}$ contains the indices of the initially prompted frames. These initial prompts identify the object of interest and are required to initialise the promptable segmentation model; they are therefore outside the corrective-prompting decision considered by our framework.

Given the initial prompts, a promptable video segmentation model s produces the initial propagated segmentation $\widehat{y} = s\,(v, \{p_i \mid i \in \mathcal{F}_{\text{init}}\})$. The propagation $\widehat{y}$ plays the role of the machine prediction in the learning-to-defer formulation. The set of unprompted frame indices eligible for corrective prompting is $\mathcal{E} = \{1, \ldots, N\} \setminus \mathcal{F}_{\text{init}}$.

3.2. Cost-aware multi-frame deferral

Given the video v and its initial propagation $\widehat{y}$, the deferral policy determines whether corrective prompts are required and, if so, how many should be requested. Let K denote the maximum number of additional frame prompts that may be requested for a video, and assume that $|\mathcal{E}| \geq K$. The policy selects $r(v, \widehat{y}) \in \{0, \ldots, K\}$, where $r(v, \widehat{y}) = 0$ retains the initial propagation and $r(v, \widehat{y}) = k > 0$ requests prompts on k additional frames.

For each $k > 0$, let $\mathcal{F}_k \subseteq \mathcal{E}$ denote the selected set of correction-frame indices, where $|\mathcal{F}_k| = k$. After obtaining the corresponding expert prompts, the resulting segmentation is $\widehat{y}_k = s\,(v, \{p_i \mid i \in \mathcal{F}_{\text{init}} \cup \mathcal{F}_k\})$.

Let $\ell(y, \widehat{y})$ denote the video-level segmentation loss. The

cost-aware multi-frame deferral objective is

$$\mathcal{L}_{\text{MF-L2D}} = \mathbf{1}[r(v,\widehat{y}) = 0]\,\ell(y,\widehat{y}) + \sum_{k=1}^{K} \mathbf{1}[r(v,\widehat{y}) = k]\,[\ell(y,\widehat{y}_k) + \alpha k]\,, \tag{2}$$

where α is the cost of one additional expert prompt. For $k > 0$, the reduction from $\ell(y,\widehat{y})$ to $\ell(y,\widehat{y}_k)$ represents the segmentation gain obtained from correction, while αk represents the corresponding expert cost.

We implement this decision using a cardinality model and a multi-frame selection model. The cardinality model determines whether the initial propagation should be retained or predicts the number of correction frames to request. Given a positive cardinality, the multi-frame selection model selects the corresponding frames from $\mathcal{E}$.

3.3. Cardinality model

The cardinality model implements the deferral policy r. Given the video and its initial propagation, it outputs $K+1$ scores and assigns a probability $\Pr_\phi(k \mid v,\widehat{y})$ to each action $k \in \{0,\dots,K\}$. The action $k = 0$ retains the initial propagation, whereas $k > 0$ specifies the number of frames to be selected by the multi-frame selection model.

At inference, the predicted cardinality is $\widehat{k} = r(v,\widehat{y}) = \arg\max_{k\in\{0,\dots,K\}} \Pr_\phi(k \mid v,\widehat{y})$. If $\widehat{k} = 0$, no further expert prompt is requested. Otherwise, the multi-frame selection model selects $\mathcal{F}_{\widehat{k}}$, and the expert annotations on these frames are applied together to update the segmentation throughout the video.

3.4. Multi-frame selection model

For a specified cardinality $k > 0$, the multi-frame selection model selects k frame indices from $\mathcal{E}$ over $q = 1,\dots,k$ selection steps. This sequential process allows each decision to account for previously selected frames, avoiding redundant choices without explicitly enumerating every possible subset. The sequentiality is over the selected-set size rather than video time, and all remaining eligible frames are considered at each step.

Throughout these selection steps, the segmentation remains $\widehat{y}$. Previously selected frames are added to the selection context, but they are not yet annotated, and the segmentation model is not re-propagated. Thus, all k frames are selected before their annotations are requested from the expert.

An encoder produces a representation $\mathbf{h}_i$ for every frame $i \in \{1,\dots,N\}$ using the video frame x_i and its initially propagated mask $\widehat{y}_i$. At selection step q, let $\mathcal{F}_{q-1} = \{i_1,\dots,i_{q-1}\}$ contain the frame indices selected at the preceding steps. The corresponding context is $\mathcal{C}_{q-1} = \mathcal{F}_{\text{init}} \cup \mathcal{F}_{q-1}$. At the first selection step ($q = 1$), $\mathcal{F}_0 = \varnothing$, and hence $\mathcal{C}_0 = \mathcal{F}_{\text{init}}$. This context allows each candidate to be evaluated by the information it adds to the frames already selected.

For each remaining candidate $i \in \mathcal{E} \setminus \mathcal{F}_{q-1}$, we apply soft cross-attention, using the candidate representation as the query and the context-frame representations as the keys and values. This produces a candidate-specific context representation $\mathbf{m}_{i,q} = \text{attn}\,(\mathbf{h}_i, \{\mathbf{h}_j \mid j \in \mathcal{C}_{q-1}\})$. Each supported cardinality k also has a learned embedding $\mathbf{e}_k$, allowing the scoring function to adapt its frame preferences to the requested set size. The candidate score is $z_{i,q}^{(k)} = \text{score}_\theta\,([\mathbf{h}_i; \mathbf{m}_{i,q}; \mathbf{e}_k])$. The attention and scoring functions are shared across all candidates, selection steps, and supported cardinalities.

The conditional probability of selecting candidate i at step q is $\Pr_\theta\,(i_q = i \mid v,\widehat{y},\mathcal{C}_{q-1},k) = \dfrac{\exp\left(z_{i,q}^{(k)}\right)}{\sum_{j\in\mathcal{E}\setminus\mathcal{F}_{q-1}} \exp\left(z_{j,q}^{(k)}\right)}$. During training, i_q is drawn by categorical sampling from this distribution. During inference, the candidate with the highest probability is selected. The partial set is then updated as $\mathcal{F}_q = \mathcal{F}_{q-1} \cup \{i_q\}$, and the context-dependent representations and candidate scores are recalculated for the next step. The original frame representations $\{\mathbf{h}_i\}_{i=1}^N$ remain fixed throughout this process.

After k steps, the sequence $\pi_k = (i_1,\dots,i_k)$ records the selection order, while the final output is the frame-index set $\mathcal{F}_k = \{i_1,\dots,i_k\} \subseteq \mathcal{E}$. The probability of the selection trajectory is

$$\Pr_\theta\,(\pi_k \mid v,\widehat{y},k) = \prod_{q=1}^{k} \Pr_\theta\,(i_q \mid v,\widehat{y},\mathcal{C}_{q-1},k)\,. \tag{3}$$

The sequence π_k is used to factorise the selection probability during training; the final decision is the unordered frame set $\mathcal{F}_k$.

3.5. Training objectives

Reward-Based Multi-Frame Selection. A video may contain multiple useful correction-frame sets, so the multi-frame selection model is not trained to reproduce a single predefined target set. Instead, it is trained using the segmentation loss obtained after prompting the selected frames.

For each cardinality k, we sample B alternative frame-index sets $\mathcal{R}_1,\dots,\mathcal{R}_B$ from $\mathcal{E}$, where $|\mathcal{R}_b| = k$ and $\mathcal{R}_b \neq \mathcal{F}_k$. Prompting the frames in $\mathcal{R}_b$ produces $\widehat{y}_{\mathcal{R}_b} = s\,(v,\{p_i \mid i \in \mathcal{F}_{\text{init}} \cup \mathcal{R}_b\})$. The lowest segmentation loss among these sampled alternatives is $b_k = \min_{1\le b\le B} \ell\,(y,\widehat{y}_{\mathcal{R}_b})$, and the advantage of the model-selected set is $a_k = b_k - \ell(y,\widehat{y}_k)$. The frame-selection loss for cardinality k is

$$\mathcal{L}_{\text{FS}}^{(k)} = -a_k \log \Pr_\theta\,(\pi_k \mid v,\widehat{y},k)\,. \tag{4}$$

Table 1. Comparison of variable-cardinality deferral across four datasets at $\alpha = 0.01$. We report segmentation quality (mIoU ↑), mean selected cardinality ($\widehat{k}$ ↓), and mean system cost ($c_{\widehat{k}}$ ↓), where $c_{\widehat{k}} = (1 - \text{mIoU}) + \alpha\widehat{k}$. Under $\mathcal{K}$, $\{0\}$, $\{0, 1\}$, and $\{0{:}4\}$ denote the allowed correction cardinalities. Rand., Single, Online, Indep., and Multi. denote random-subset selection, single-frame selection, online triggering, independent ranking, and multi-frame selection, respectively.

Method	k	Sel.	VOST			CAMotion			SUN-SEG			VTUS		
			mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓
No correction	0	–	.473	0.00	.527	.802	0.00	.198	.711	0.00	.289	.627	0.00	.373
Random	0:4	Rand.	$.587_{\pm.005}$	2.00	.433	$.819_{\pm.003}$	2.00	.191	$.762_{\pm.008}$	2.00	.248	$.694_{\pm.004}$	2.00	.316
L2RP [13]	0,1	Single	$.602_{\pm.002}$	0.76	.406	$.822_{\pm.002}$	0.47	.180	$.774_{\pm.006}$	0.52	.229	$.722_{\pm.001}$	0.88	.282
Lazy-XMem [18]	0:4	Online	.644	3.33	.390	.845	3.11	.171	.830	3.65	.188	.764	4.00	.256
LECODU [23]	0:4	Rand.	$.658_{\pm.001}$	3.21	.374	$.837_{\pm.002}$	1.28	.169	$.836_{\pm.004}$	2.01	.174	$.755_{\pm.002}$	3.26	.261
Top-$k(x)$ [8]	0:4	Indep.	$.662_{\pm.005}$	2.82	.366	$.838_{\pm.002}$	1.30	.168	$.842_{\pm.005}$	1.80	.167	$.795_{\pm.005}$	2.90	.220
Ours	**0:4**	**Multi.**	$\mathbf{.699}_{\pm.002}$	2.56	**.327**	$\mathbf{.847}_{\pm.001}$	1.19	**.159**	$\mathbf{.850}_{\pm.003}$	1.71	**.158**	$\mathbf{.819}_{\pm.002}$	2.81	**.195**
Oracle	*0:4*	*Best*	.714	2.86	.314	.851	1.10	.155	.874	1.84	.135	.831	2.87	.183

A positive advantage indicates that the selected set produces a lower segmentation loss than the sampled alternatives and therefore increases its selection probability. A negative advantage reduces its probability.

The multi-frame selection model is trained across all supported cardinalities $k \in \{1, \ldots, K\}$, with $\mathcal{L}_{\text{FS}} = \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_{\text{FS}}^{(k)}$. The frozen segmentation model is executed within the training loop to generate the post-correction masks and rewards. For each cardinality, it evaluates the selected set and B sampled alternatives, requiring $K(B + 1)$ online propagations per video update without enumerating all possible frame subsets.

Cost-Aware Cardinality Training. The multi-frame selection model is trained first and then frozen. During cardinality-model training, the frozen selection model produces one frame set $\mathcal{F}_k$ for each supported cardinality $k > 0$. For every cardinality $k \in \{0, \ldots, K\}$, we compute the corresponding system cost as

$$c_k = \begin{cases} \ell(y, \widehat{y}), & k = 0, \\ \ell(y, \widehat{y}_k) + \alpha k, & k > 0. \end{cases} \tag{5}$$

Thus, c_0 is the system cost of retaining the initial propagation, whereas c_k for $k > 0$ additionally includes the cost of requesting k expert prompts. The cardinality targets therefore reflect the trade-off between segmentation quality and expert effort.

To make the objective in Equation (2) trainable, we adapt a cost-weighted multiclass surrogate loss [1]. The action costs are normalised independently for each video to obtain $\widetilde{c}_k \in [0, 1]$, and the cardinality loss is

$$\mathcal{L}_{\text{card}} = -\sum_{k=0}^{K} (1 - \widetilde{c}_k) \log \Pr_{\phi}(k \mid v, \widehat{y}). \tag{6}$$

Actions with lower system cost receive larger weights, allowing the model to learn the relative quality–cost trade-off among all supported cardinalities.

During cardinality model training, the frozen segmentation model evaluates one selected set for each positive cardinality, requiring K online propagations per video update. Neither training stage enumerates the combinatorial set of possible frame subsets. At inference, the cardinality model predicts one $\widehat{k}$ and the selector chooses all $\widehat{k}$ frames without intermediate segmentation reruns. Once their expert prompts are available, SAM2 incorporates them through a single corrected propagation.

4. Experiments

We evaluate the proposed framework by posing three questions. **Q1**, Does the variable-cardinality deferral improve the trade-off between segmentation quality and expert cost? **Q2**, Does the proposed multi-frame selection model identify better correction frames than existing frame-selection methods under a fixed correction budget? **Q3**, Does the cardinality model allocate expert effort to videos that benefit most from additional corrections?

4.1. Experimental setup

Datasets and prompting protocol. We evaluate on VOST [15], CAMotion [21], SUN-SEG [5], and VTUS [19], using the official train/test splits. We define ten uniformly spaced temporal locations per video: the first receives the initial ground-truth mask prompt, and the remaining nine are correction candidates. These locations restrict only prompt placement; SAM2 propagates over the complete video, and training rewards and video-level mIoU are computed over all frames. Unless otherwise stated, corrective prompts are ground-truth masks, and we allow up to $K = 4$ additional corrections. This setting allows the method to address multiple propagation failures while lim-

Table 2. Fixed-cardinality frame selection at $k = 3$. Rounds column shows 1 when all three frames are selected in a single round, 3 when one frame is selected per round.

Method	Rounds	VOST	CAMotion	SUN-SEG	VTUS
Random frames	1	0.613	0.831	0.787	0.727
Min-IoU frames	1	0.647	0.843	0.791	0.748
HVSA [11]	1	0.663	0.844	0.828	0.781
IVOS-W [22]	3	0.655	0.845	0.818	0.769
EVA-VOS [2]	3	0.667	0.848	0.825	0.786
Top-k [8]	1	0.670	0.847	0.829	0.772
Ours	**1**	**0.688**	**0.852**	**0.862**	**0.799**
Oracle	*1*	*0.712*	*0.859*	*0.875*	*0.828*

iting the maximum expert effort. Additional details and results for other values of K are provided in the appendix.

Implementation details. We use a frozen SAM2.1 Hiera-Tiny model [14] as the promptable video segmentation model. For each sampled frame, we use the embeddings produced by the SAM2 memory encoder, which jointly encode the frame appearance and its corresponding current predicted mask. These representations are processed by independently trained spatial-temporal encoders for the multi-frame selection model and cardinality model, followed by MLP decision heads. The frame selector additionally uses candidate-to-context attention and learned cardinality embeddings.

The frame selector is trained for 400 epochs with $B = 4$, after which it is frozen and the cardinality model is trained for 200 epochs. Both models use AdamW (learning rate 10^{-5}, weight decay 10^{-4}, and batch size 16) and are trained on an NVIDIA GeForce RTX 5090 GPU. All learned methods are evaluated over three random seeds. Reported $\pm$ values denote the standard deviation across these runs. Additional implementation details are provided in the appendix.

Evaluation metrics. We report video-level mean intersection over union (mIoU), average cardinality, and the average system cost, which combines segmentation loss with expert prompting cost. Unless otherwise stated, all methods use $\alpha = 0.01$; results for additional α values are in the appendix.

Evaluation protocols and comparison methods. To evaluate the quality–cost trade-off (Q1), we allow each method to request $k \in \{0, \dots, 4\}$ corrections and report mIoU, mean selected cardinality, and system cost. We compare against a no-correction reference, a random policy, and four baselines. Lazy-XMem [18] triggers corrections online using per-frame uncertainty, while L2RP [13] supports only $k \in \{0, 1\}$. We additionally adapt two variable-cardinality multiple-expert deferral methods from classification by treating candidate frames as expert actions. LECODU [23] predicts how many corrections to request and samples that number of frames at random. Adaptive Top-$k(x)$ [8] predicts the correction cardinality and selects the highest-ranked frames according to their independently estimated correction utility. We also report an oracle upper bound that, for each video, jointly selects the cardinality and corresponding frame set that yield the lowest system cost.

We also evaluate performance across non-deferral coverage levels. For each method, using its own output probabilities, we threshold $d = \max_{k>0} \Pr_\phi(k \mid v, \widehat{y}) - \Pr_\phi(0 \mid v, \widehat{y})$; videos above the threshold are corrected using the most probable positive cardinality, while the remainder retain the current propagation. Sweeping the threshold yields the mIoU–coverage curve.

To compare frame selection independently of the cardinality (Q2), we fix the number of corrections per video and report the resulting mIoU. The multi-frame selection model is conditioned on the specified cardinality and directly selects the corresponding number of frames. We compare against random selection, a min-IoU heuristic that selects the k frames with the lowest IoU between the initial propagation and ground truth, Top-k ranking, and adaptations of HVSA [11], EVA-VOS [2], and IVOS-W [22]. HVSA and Top-k perform batch frame selection, whereas EVA-VOS and IVOS-W select frames sequentially, re-propagating after each correction before selecting the next frame. We also report an oracle upper bound that, for each video and fixed k, selects the frame subset that yields the highest mIoU.

To analyse the cardinality model (Q3), we group videos by their initial video-level mIoU and examine both the mean predicted correction cardinality and the resulting improvement in segmentation performance for each group.

Further details of the baseline adaptations and evaluation protocols are provided in the appendix.

5. Results and analysis

Quality–cost trade-off (Q1). Table 1 shows that, without correction, the initial propagation yields the lowest mIoU, while random correction improves it. L2RP requests the fewest corrections but is limited to a single frame, whereas Lazy-XMem requests 3–4 corrections. LECODU and Top-$k(x)$ incur similar system costs on CAMotion, while Top-$k(x)$ achieves lower costs on VOST, SUN-SEG and VTUS. Our method achieves the highest mIoU with the lowest system cost. Compared with Top-$k(x)$, ours reduces the mean cardinality by 3.1–9.2% and the system cost by 5.4–11.4%. Although the oracle defines the upper bound, ours consistently remains the closest reducing 28.1–75.0% of the system-cost gap between Top-$k(x)$ and the oracle.

Figure 3 further shows that our method achieves the highest mIoU across non-deferral coverage levels, show-

Table 3. Ablation on the number of sampled frame subsets B used to construct the reward baseline during training. Values are video-level mIoU.

Dataset	$B = 0$	$B = 1$	$B = 2$	$B = 4$
VOST	0.665	0.676	0.681	**0.699**
CAMotion	0.825	0.830	0.832	**0.847**
SUN-SEG	0.826	0.836	**0.851**	0.850
VTUS	0.787	0.800	0.802	**0.819**

Table 4. Ablation of the cardinality-decision and frame-selection components. "Cardinality" denotes the cardinality model, while "Multi-frame" denotes the multi-frame selection model. Values are video-level mIoU.

Cardinality decision	Frame selection	VOST	CAMotion	SUN-SEG	VTUS
Random	Multi-frame	0.621	0.829	0.802	0.746
Cardinality	Random	0.674	0.831	0.834	0.806
Cardinality	**Multi-frame**	**0.699**	**0.847**	**0.850**	**0.819**

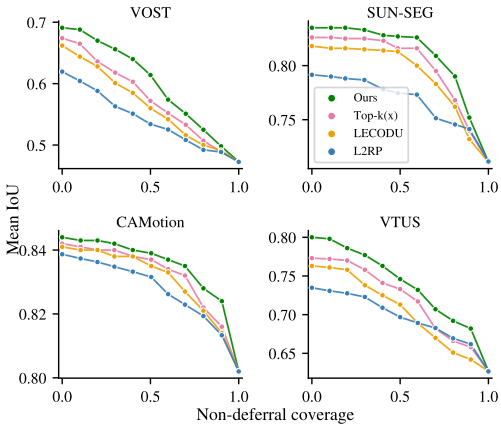


Figure 3. Mean IoU at different non-deferral coverage levels. Coverage is the fraction of videos that retain the initial segmentation without corrective prompting: 0 means that all videos are corrected, whereas 1 means that none are corrected. Higher curves indicate better segmentation quality at the same coverage.

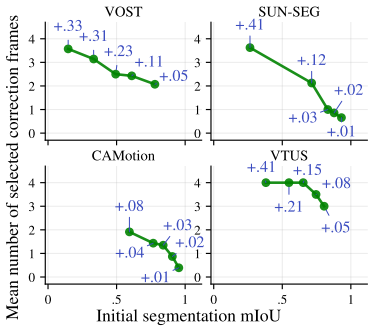


Figure 4. Mean number of selected correction frames versus initial segmentation quality. Videos are grouped into five initial-mIoU bins, and the labels show the mean mIoU gain after correction. Lower-quality segmentations receive more corrections and larger gains, indicating that the policy allocates expert effort where it is most useful.

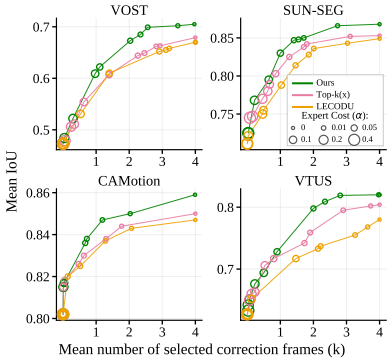


Figure 5. Quality–cost trade-off under varying expert-cost weight α. Each point reports the video-level mIoU and mean selected cardinality for one value of α, with marker size increasing with α. Our method is generally above the baselines at comparable correction budgets

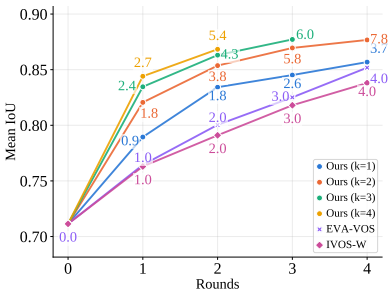


Figure 6. Mean IoU over corrective-prompting iterations on SUN-SEG at $\alpha = 0.001$. Our models with maximum cardinality $K \in \{1, 2, 3, 4\}$ select correction frames in batches, whereas EVA-VOS and IVOS-W select one frame per iteration. After each iteration, the selected frames are prompted and the video is re-propagated. Point labels show the cumulative number of selected frames.

ing its performance gains are consistent across wide range of operating points rather than tied to a particular decision threshold.

These results show that variable-cardinality deferral improves the quality–cost trade-off by allocating correction effort according to each video, achieving higher segmentation quality without requesting more expert prompts.

Frame selection at fixed cardinality (Q2). Table 2 compares frame selection methods with a fixed expert budget of $k = 3$. IVOS-W and EVA-VOS perform sequential frame selection, re-propagating after each correction, whereas the batch methods select all frames in a single batch from the initial segmentation. Despite not benefiting from intermediate propagation updates, our method achieves the highest non-oracle mIoU on all four datasets improving over EVA-VOS by 0.5–4.5% and batch Top-k by 0.6–4.0%. Results

for $k \in 1, 2, 3, 4$, reported in the appendix, show the same trend.

Allocation of expert effort (Q3). Figure 4 shows that the policy allocates larger correction budgets to videos with lower initial mIoU and fewer corrections to those with higher initial mIoU. The mIoU gains are positive across all bins and are generally larger for lower-quality initial propagations. These results indicate that the policy allocates expert effort according to both the current propagation quality and the expected benefit of additional corrections.

6. Ablation studies

Effect of the expert-cost weight: Figure 5 shows that increasing α reduces both the mean selected cardinality and mIoU. Our method generally achieves higher mIoU than the baselines at comparable correction budgets, showing a consistent quality–cost trade-off across datasets.

Number of sampled frame subsets: Table 3 shows that increasing the number of sampled subsets generally improves mIoU. Compared with no sampled alternatives ($B = 0$), $B = 4$ improves mIoU by 0.034 on VOST, 0.022 on CAMotion, and 0.032 on VTUS. SUN-SEG performs best at $B = 2$, with little change at $B = 4$. We therefore use $B = 4$ in the main experiments.

Cardinality and frame-selection components: Table 4 shows that replacing either learned component with a random alternative reduces video-level mIoU. Randomising the cardinality decision causes the larger reduction, particularly on VOST and VTUS, while random frame selection also lowers performance across all datasets. Combining the learned cardinality model with multi-frame selection achieves the highest mIoU on every dataset.

Multiple correction rounds: Although our framework requests all corrections in a single round, the same policy can be reapplied after each re-propagation. Figure 6 reports this setting on SUN-SEG for maximum cardinalities $K \in \{1, 2, 3, 4\}$. Larger batches reach a given mean IoU in fewer interaction rounds, which matters because each round returns the video to the expert for review. Each round applies the policy to the current propagation alone, without distributing a correction budget across rounds, and optimising jointly across rounds is a natural extension.

7. Conclusion

We formulated corrective prompting for promptable video segmentation as a variable-cardinality learning-to-defer problem that accounts for how the value of each correction depends on the other selected frames. Our framework couples a cardinality model that determines whether and how many corrections are warranted under a given expert cost with a multi-frame selection model that conditions each choice on the frames already selected and requests the resulting corrections in a single round. Experiments with a SAM2 model across four visual and clinical datasets show that our method achieves the highest mIoU and lowest system cost among non-oracle methods while requesting fewer corrections than the strongest baseline. These findings show that coupling cost-aware cardinality decisions with multi-frame selection improves the quality–cost trade-off while consolidating corrective prompting into a single round.

LLM Usage Statement

OpenAI Codex was used to assist with migrating the existing manuscript and supplementary material from the AAAI template to the CVPR template, adapting LaTeX formatting, and checking compilation and layout.

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